

Then, choose the reference resistor and IDAC current value. Ideally, choosing the largest IDAC current provides the best performance by increasing the sensor signal above any noise in the system. However, there are several other considerations in determining the values. First, higher current may lead to self-heating of the RTD, which adds error to the measurement. Second, the reference resistance acts as a level shift for the sensor measurement. This level shift is used to raise the DC bias of the analog input signal so that the voltage is within the input range of the PGA. Generally, the analog input signal is set near mid-supply for best operation.

To verify that the design is within the ADC range of operation, Calculate the voltages for AIN1 and AIN2 and the maximum differential input voltage. Verify that V_{AIN1} and V_{AIN2} are within the input range of the PGA given the gain setting and supply voltage. Use the maximum RTD resistance based on the desired temperature measurement.

$$V_{AIN1} = I_{IDAC1} \cdot (R_{RTD} + R_{REF}) \quad (15)$$

$$V_{AIN2} = I_{IDAC1} \cdot R_{REF} \quad (16)$$

Additionally, the output voltage of the IDAC source calculated from V_{AIN1} must be low enough from AVDD to be within the compliance voltage of the IDAC current source. When the IDAC output voltage rises too close to AVDD, the IDAC loses compliance and the excitation current is reduced.

The reference resistor, R_{REF} must be a precision resistor with high accuracy and low drift. Any error in the R_{REF} reflects the same error in the RTD measurement. The REFP0 and REFN0 pins are shown connecting to the R_{REF} resistor as a Kelvin connection to get the best measurement of the reference voltage. This eliminates any series resistance as an error from the reference resistance measurement.

The lead wire resistance is an error term in the two-wire RTD measurement. The previous calculations neglect the lead resistances, but can be added to the R_{RTD} term.

2.1.4 Measurement Conversion

$$\text{Output Code} = 2^{23} \cdot \text{Gain} \cdot V_{RTD} / V_{REF} = 2^{23} \cdot \text{Gain} \cdot (I_{IDAC1} \cdot R_{RTD}) / (I_{IDAC1} \cdot R_{REF}) = 2^{23} \cdot \text{Gain} \cdot R_{RTD} / R_{REF} \quad (17)$$

$$R_{RTD} = R_{REF} \cdot \text{Output Code} / (2^{23} \cdot \text{Gain}) \quad (18)$$

2.1.5 Generic Register Settings

- Select multiplexer settings for AINP and AINN to measure leads 1 and 2 of the RTD
- Enable the PGA, set gain to desired value
- Select data rate and digital filter settings
- Select reference input to measure R_{REF} for ratiometric measurement
- Enable the internal reference (the IDAC requires an enabled internal reference)
- Set IDAC magnitude and select IDAC1 output pin to drive lead 1 of the RTD